

Emanuel Umbelino

Software Engineer | Full-Stack | AI & Automation Specialist

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SUMMARY

Software engineer with 10+ years of experience, programming since age 15. Full-stack with a strong front-end, UX and interface background, specialized in AI agents, LLM-powered automation and AI-assisted Spec-Driven Development (SDD). Delivered production systems for Petrobras, Shell, Petronas, CNOOC, Samsung and Carrefour. Fully remote since 2019, working across time zones with international clients. Product-minded: I focus on understanding client needs to deliver the right solution.

PROFESSIONAL EXPERIENCE

PUC-Rio - GTEP | Oil & Gas Industry

Nov 2019 - Present | Remote

Full-Stack Developer

- Build web platforms for well data processing and analysis for Petrobras, Petronas, Shell and CNOOC, in multi-million (BRL) projects.
- Own delivery end to end: UX prototyping, front-end, backend and database, with clean, dynamic interfaces and clear user guidance.
- Used Claude with Spec-Driven Development (SDD) to accelerate platform development, and automated CI/CD pipelines with Docker.

VONEX.AI | AI Customer Service & Automation

Aug 2022 - Sep 2026 | Remote

AI Specialist & Senior Front-End Engineer (Front-End Team Lead)

- Built an omnichannel customer service platform with AI for Samsung, Carrefour, Jequití, Consigaz and others, handling thousands of messages daily.
- Integrated Google Gemini and OpenAI (ChatGPT) APIs into 24/7 automated support flows with AI agents, delivering complete and personalized service to each customer.
- Delivered advertising campaign dispatch with thousands of sends.
- Led the front-end team for a period.

IBM | Oil & Gas Industry

Sep 2018 - Oct 2019 | On-site

UX Research Intern - Front-End

- Built interfaces, optimized performance and integrated the web application with backend services.
- Worked with domain specialists to turn designed screens into a functional application.

OTHER PROJECTS

ShopList | Mobile & Web App

May 2026

- Designed and built end to end a collaborative shopping list app: lists shared by an 8-character code, no sign-up, with real-time sync across devices via WebSocket.
- Installable PWA with offline resilience (edits are retried and merged on reconnection), dark mode and i18n (PT, EN, ES).
- REST API documented with Swagger, with rate limiting, scheduled jobs (list expiry, stats aggregation) and a usage dashboard.
- Unit, integration and E2E tests; CI/CD with GitHub Actions and Docker, deployed on Railway.

Live: shop-list.up.railway.app/

SAGIL | Mobile App

Jul 2024 - Oct 2024

- Built the front-end of an IoT app for medical refrigerators: users register refrigerators, configure them, connect them to the network and integrate them with a dashboard and controller for temperature monitoring.

Leite Sobre Rodas (Fiocruz) | Mobile & Web App

Sep 2018 - Nov 2019

- Built an app managing the full breast milk donation flow, connecting donors to a hospital.
- Included a delivery-style module with GPS and route optimization, so a collector picks up milk at donors' homes and delivers it to Fiocruz.

SKILLS

AI: Gemini API, OpenAI API, LLM Agents, Prompt Orchestration, RAG Systems, Spec-Driven Development (SDD), Claude Code, Cursor

Languages: TypeScript, JavaScript, Python, C#, Dart, Java, C++

Front-End: Angular, React, Vue, Flutter, Vite, TanStack Query, Tailwind CSS, SCSS, HTML, CSS

Backend: .NET, Node.js, Fastify, Django, Laravel, RESTful APIs, Socket.IO

Data & DevOps: PostgreSQL, MongoDB, Redis, Drizzle ORM, Docker, CI/CD, GitHub Actions

Testing: Vitest, Playwright, Testing Library

Design: Figma, UX/UI, Design Thinking, Prototyping

Soft Skills: Clear async communication, Self-management, Product mindset, Fast learning, Team leadership

EDUCATION

Pontifical Catholic University of Rio de Janeiro (PUC-Rio)

Feb 2017 - Jul 2022

B.Sc. in Computer Science

NAVE - CEJLL

Feb 2014 - Dec 2016

Technical Degree in Digital Game Programming

LANGUAGES

Portuguese (Native) | English (Advanced)